Parameter-Efficient Local-Context Cooperation via Vision Foundation Models for UHR Remote Sensing Image Segmentation

Qi Li, Chunxiao Chen, Jiaxin Cai, Junjie Huang, Yansheng Li, *Senior Member, IEEE*, and Wenxi Liu, *Senior Member, IEEE*

***Abstract*—Ultrahigh resolution (UHR) remote sensing image segmentation aims to achieve a fine-grained understanding of complex ground scenes. In recent years, vision foundation models (VFMs) have shown strong capability in learning generic structural priors from large-scale visual data, indicating great potential for such fine-grained scene understanding. However, their application to UHR remote sensing images remains limited, as the massive parameter scales of VFMs are difficult to train under UHR remote sensing images. Motivated by the success of the parameter-efficient fine-tuning paradigm on VFMs, we propose a novel parameter-efficient local-context cooperation (PEACE) framework, which significantly reduces trainable parameter overhead while improving segmentation accuracy. In particular, PEACE leverages a shared VFM with minimal trainable parameters to collaboratively process local and corresponding contextual patches partitioned from the UHR remote sensing image. A multireceptive local adapter (MRLA) and a multireceptive context adapter (MRCA) are designed to capture spatial features of local and contextual inputs across multiple receptive fields. Finally, contextual semantics are integrated into local representations. Furthermore, a context-sensitive assistance strategy (CSAS) leverages the correct prediction of the context to effectively overcome the primary limitations of the patch-based training paradigm. Experimental results demonstrate that PEACE effectively exploits VFMs and remote sensing foundation models (RSFMs) with minimal parameter increments and achieves versatility and superior performance across several UHR remote sensing image benchmarks.**

***Index Terms*—Parameter-efficient fine-tuning, remote sensing image, semantic segmentation, ultrahigh resolution (UHR), vision foundation models (VFMs).**

I. Introduction

With the rapid advancement of sensor technology, acquiring ultrahigh resolution (UHR) remote sensing images has become increasingly feasible. These images, rich in detail and complex scene information, have been widely applied in fields such as urban planning [1], [2], land cover classification [3], [4], and water resource monitoring [5], [6]. To better analyze these images, UHR remote sensing image segmentation provides powerful support by enabling pixel-level semantic classification.

In recent years, many approaches [7], [8], [9] for UHR remote sensing image segmentation have adopted the design paradigm of deep-shallow networks. The deep network processes downsampled images to extract global features, while the shallow network directly handles the original UHR images to preserve detailed information. The final segmentation is achieved by fusing the outputs from both networks. However, as the remote sensing image resolution scales up further (e.g., as high as 12K $\times$ 12K pixels), these methods suffer from excessive GPU memory consumption and discard fine object boundaries and thin structures, leading to significant performance degradation and making them unsuitable for real-world applications.

To address this issue, recent methods [10], [11] have explored patch-based inference combined with contextual information. Although effective, the patch-based frameworks have notable drawbacks, making them incompatible with numerous applications. On the one hand, these approaches typically adopt multiencoder designs with full fine-tuning, which impose significant parameter storage burdens and complicate deployment transfers. On the other hand, such designs commonly employ vanilla vision representations, overlooking the potential of pretrained knowledge from large-scale vision foundation models (VFMs). Their generic structural priors capture typical object shapes, surface patterns, and spatial context, which transfer well to complex UHR remote sensing scenes and provide rich structural cues for fine-grained segmentation.

Inspired by the recent success of the parameter-efficient fine-tuning paradigm on VFMs [12], [13], we propose a novel parameter-efficient local-context cooperation (PEACE) framework, which effectively reduces trainable parameters while improving segmentation performance in UHR remote sensing scenarios. The core idea is to adaptively process local and corresponding contextual patches partitioned from the UHR remote sensing image using a vision foundation backbone with minimal incremental parameter adjustments, enabling collaboration between local features and contextual information. In particular, the network of PEACE is built on a VFM and consists of three cooperative components, including Local-Net, Context-Net, and Injection-Net. Local-Net and Context-Net share the same pretrained back-

Received 9 January 2026; revised 27 April 2026 and 4 June 2026; accepted 30 June 2026. Date of publication 6 July 2026; date of current version 13 July 2026. This work was supported in part by the Innovation of Policing Science and Technology, Fujian Province under Grant 2024Y0061. *(Corresponding author: Wenxi Liu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3710197

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bone and are differentiated only by lightweight adapters, which specialize the VFM features for local and contextual inputs, respectively. This dual adapter design enables PEACE to reuse generic structural priors from VFMs while tailoring them to both fine details and large-scale context in UHR remote sensing scenes. The Injection-Net produces cooperative features by injecting contextual semantics from Context-Net into local features from Local-Net, addressing the limitations of narrow local receptive fields.

To enable more flexible and robust adaptation of VFMs to objects of varying scales in UHR remote sensing images, we introduce two lightweight adapters: a multireceptive local adapter (MRLA) for Local-Net and a multireceptive context adapter (MRCA) for Context-Net. MRLA follows a conventional multireceptive design with depthwise convolutions to densely capture spatial structures across different scales, refining fine details in local patches. On the contrary, MRCA employs multireceptive depthwise dilated convolutions to perform sparse sampling, effectively skipping redundant information and mitigating the excessive response of erroneous information, thereby focusing on critical contextual cues. This asymmetric design differs from previous multireceptive modules, explicitly tailoring a local adapter for dense modeling and a context adapter for sparse sampling.

Although the proposed framework efficiently absorbs both local and contextual information, existing training paradigms fail to explicitly capture the heterogeneous contributions of different local regions to semantic discrimination. In UHR local patches, traditional uniform optimization strategies weaken the model's sensitivity to large-scale regions (e.g., water bodies and forests) due to the narrow field of view. To address this, we propose a context-sensitive assistance strategy (CSAS) leveraging the distinct regions of interest for Local-Net and Context-Net to overcome the primary limitations of the patch-based training paradigm. In particular, the CSAS uses the correct predictions from Context-Net under large receptive fields. These regions reflect the semantic understanding of broad regions by the Context-Net, guiding the Local-Net to perceive such regions.

We evaluate our framework on several public UHR remote sensing benchmarks, including Five-Billion-Pixels (7K $\times$ 7K) [3], GLH-Water (12K $\times$ 12K) [5], and Inria Aerial (5K $\times$ 5K) [1], and achieve state-of-the-art performance across all of them. To assess versatility, we adapt our framework to various VFMs, such as Swin Transformer [14], SAM2 [15], DINOv2 [16], and DINOv3 [17], with the DINO series introducing only 6.5M additional parameters over the baseline. We also evaluate generalization using remote sensing foundation models (RSFMs), including SkySense [18] and SkySense++ [19], which demonstrate notably stronger generalization. Furthermore, we validate the effectiveness of each component and analyze model efficiency.

Overall, our contributions are summarized as follows.

1) We propose a novel PEACE framework for UHR remote sensing image segmentation that is built on a VFM, effectively reusing its generic structural priors while substantially reducing trainable parameters and improving segmentation accuracy.
2) To adapt VFM features to objects at diverse scales, we design two asymmetric multireceptive adapters, MRLA and MRCA, for local and context inputs, respectively. MRLA performs dense modeling of local structures, whereas MRCA extracts key contextual semantics by sparse sampling, enabling VFMs to handle complex UHR remote sensing scenes more effectively.
3) To effectively overcome the primary limitations of the patch-based training paradigm, we propose CSAS, which leverages the correct prediction of the context to guide the local model to perceive large-scale areas.
4) Our framework alleviates the trainable parameter burden, demonstrates versatility across various existing VFMs and RSFMs, and achieves state-of-the-art performance on several public UHR remote sensing image benchmarks.

The article is structured as follows: Section II reviews related work. Section II-C details the proposed PEACE framework. Section IV presents and analyzes experimental results. Finally, Section VI concludes the study with future work.

II. Related Work

In this section, we survey the related work on UHR semantic segmentation. In addition, the relevant work on generic semantic segmentation and parameter-efficient fine-tuning for vision is briefly summarized.

A. UHR Semantic Segmentation

UHR semantic segmentation can be categorized into two main paradigms. The first paradigm directly downsamples the UHR image to a lower resolution, performs semantic inference using conventional segmentation models, and then upsamples the result back to the original resolution. Although this approach enables fast inference, it often overlooks small or elongated objects, leading to the loss of detailed information. To address this limitation, several works [5], [7], [8], [9], [20], [21], [22] simultaneously process the UHR image at both its original and downsampled resolutions. ISDNet [7] presents a relation-aware feature fusion module to fuse features from two networks. WSDNet [8] and GPWFormer [20] both employ wavelet transforms to enhance feature representations. SGHRQ [21] proposes a spatial-guided high-resolution query module to infer high-quality semantic masks. RUE [9] boosts the dual-stream architecture through resolution-biased uncertainty estimation. HLRC [22] decouples high- and low-resolution tasks to mitigate semantic inconsistency between different networks. However, these methods still suffer from severe information loss and incur substantial GPU memory consumption when the image resolution further increases. The second paradigm divides the UHR image into multiple local patches. Each patch is processed independently by a conventional segmentation model, and then all the local results are merged into a full UHR semantic mask. This approach struggles to accurately predict large-scale semantic regions because of the limited receptive field of local patches. To improve local semantic discrimination, several works [10], [11], [23], [24], [25], [26], [27], [28], [29] introduce contextual

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regions beyond the local patches. GLNet [10] incorporates global information into local features in the deep convolutional layers. MagNet [24] progressively refines coarse masks via multistage processing. FCtL++ [11], [25] further enhances local semantics by integrating multiscale context. SRMF [29] proposes multiscale anchored region sampling to obtain richer contextual information. Nevertheless, these methods incur significant parameter storage burdens due to the multiencoder designs and overlook the potential of pretrained knowledge from large-scale VFMs. In addition, some works adopt the single-branch inference framework [30], [31]. REST [30] processes UHR images directly without cropping by parallel computation. Its core innovation lies in a spatial parallel interaction mechanism that distributes features across multiple GPUs, then fuses them through feature exchange to achieve comprehensive global context modeling. However, it heavily relies on the number of available GPUs, which limits its flexibility in resources. Overall, these methods are difficult to deploy in real-world applications due to resource constraints and suboptimal performance.

B. Generic Semantic Segmentation

Early semantic segmentation approaches primarily rely on CNN-based methods [32], [33], [34], [35], aiming to expand the receptive field to capture broader contextual information. In recent years, the emergence of Transformer-based methods [14], [36], [37], [38], [39] has led to significant breakthroughs in semantic segmentation by effectively modeling long-range dependencies across the entire image. Despite these advances, most existing methods still struggle to deliver optimal results when applied to UHR remote sensing images due to limited computational resources. In addition, efficient designs [40], [41], [42], [43] have been proposed to balance accuracy and computational cost. Recently, Many studies have focused on Mamba-based visual architectures due to their ability to model long-range dependencies with linear computational complexity. RSMamba [44] offers substantial advantages in terms of representational capacity and computational efficiency, while DynamicVis [45] is specifically designed to overcome computational barriers associated with high-resolution inputs. However, their application to UHR remote sensing images remains limited due to suboptimal performance.

C. Parameter-Efficient Fine-Tuning for Vision

With the advancement of large-scale models [46], [47], [48], parameter-efficient fine-tuning (PEFT) has become increasingly significant. Originally proposed in natural language processing [49], [50], this approach has recently expanded to computer vision applications [51], [52], [53]. The key idea is to preserve the performance of the pretrained VFMs [15], [16] by fixing their weights and introducing only a small number of trainable parameters. Representative PEFT techniques include adapter-based methods [12], [49] and prompt-based approaches [13], [54]. Recently, PEFT techniques have also been explored in the remote sensing domain [55], [56]. RSPrompter [55] proposes a prompt learning framework to adapt the foundation model for remote sensing instance segmentation. Such approaches demonstrate the potential of PEFT strategies for remote sensing tasks.

III. Proposed Method

In this section, we first present an overview of our framework and then introduce its key components: Multireceptive Local Adapter, Multireceptive Context Adapter, Injection-Net, and Context-Sensitive Assistance strategy.

A. Overview

In this work, we adopt the patch-based segmentation paradigm [25] to address the challenge of UHR remote sensing image segmentation, as the excessively high resolution (e.g., as high as 12K $\times$ 12K pixels) makes it infeasible to process the entire image in one pass. In particular, given a UHR remote sensing image $U \in \mathbb{R}^{3\times H\times W}$, we partition it into $N \times M$ overlapping local patches $P_{nm} \in \mathbb{R}^{3\times H_P\times W_P}$, where $n \in \{1,\ldots,N\}$ and $m \in \{1,\ldots,M\}$, along the height and width dimensions. To mitigate the limited receptive field inherent in processing small patches, we follow previous approaches [25] by extracting contextual regions $C_{nm} \in \mathbb{R}^{3\times H_C\times W_C}$ with $H_C = \lambda H_P$ and $W_C = \lambda W_P$ (where $\lambda > 1$) to expand the effective receptive field. Each context C_{nm} is centered on the local patch P_{nm} and fully covers it. Any extension beyond the UHR remote sensing image U is zero-padded and excluded from computation. Subsequently, our proposed parameter-efficient local-context cooperation (PEACE) framework processes each local patch and its corresponding context to generate segmentation results, which effectively preserves fine-grained details while capturing contextual information. Finally, all local predictions are aggregated to form the complete UHR segmentation mask. For brevity, the local patch P_{nm} and its context C_{nm} are denoted as P and C, respectively.

As illustrated in Fig. 1, the proposed PEACE framework consists of four key components: Local-Net, Context-Net, Injection-Net, and a CSAS. Local-Net and Context-Net consist of L shared frozen Transformer layers pretrained from vision foundation model (e.g., Vision Transformer [36] layers pretrained on DINOv2 [16]), each typically comprising an attention operation and a feed-forward network (FFN). To adapt to the distinct characteristics of the local patch P and its context C, we equip Local-Net and Context-Net with the MRLA and MRCA, respectively. These adapters are inserted immediately after the attention operation and FFN in each layer. Building upon these, Injection-Net is responsible for transmitting contextual information into local patches to generate cooperative features, enhancing local semantic discriminability. During training, the CSAS leverages the output of Context-Net to guide Local-Net in attending to broad spatial regions. In Sections III-B–III-E, we describe each proposed component.

B. Multireceptive Local Adapter

For UHR remote sensing image understanding, effective feature extraction requires perceiving and integrating visual

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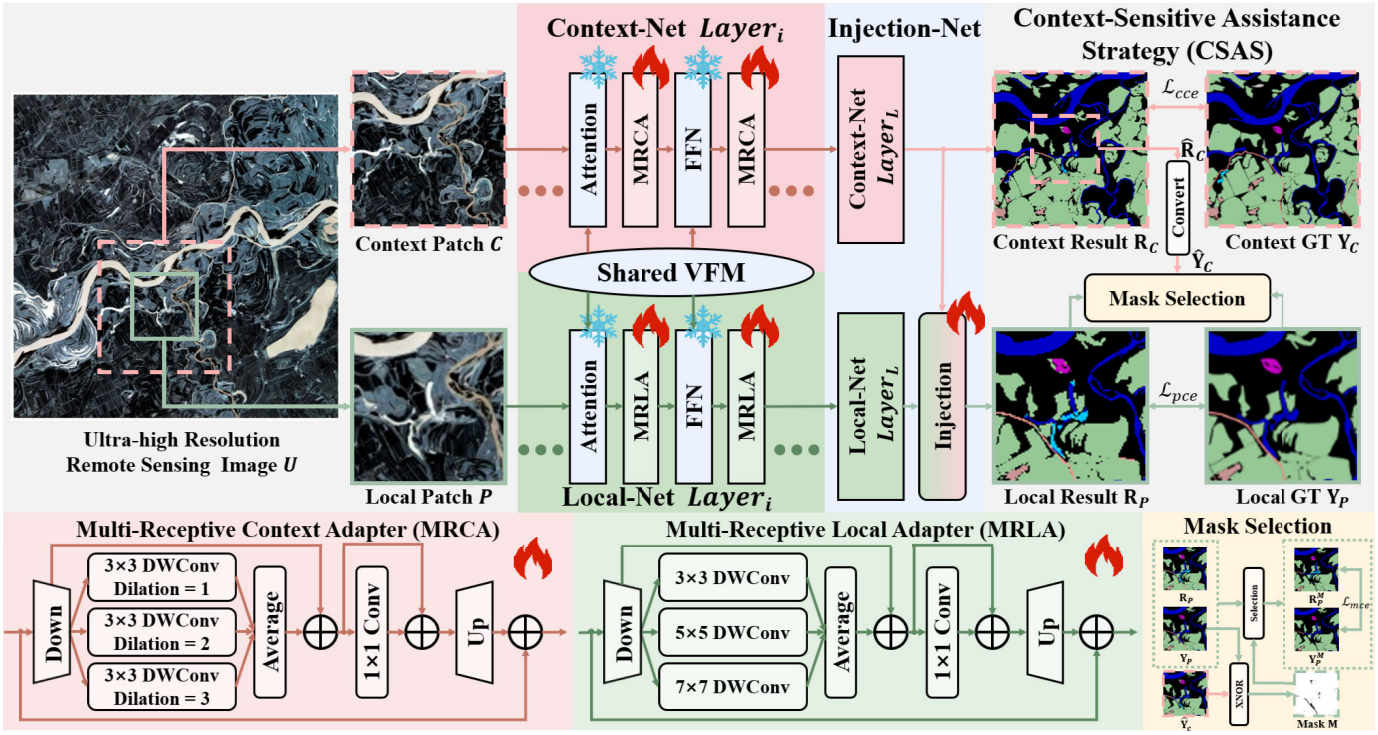


Fig. 1. Illustration of our proposed parameter-efficient local-context cooperation (PEACE) framework. In particular, a local patch cropped from the UHR remote sensing image and its corresponding context patch are fed into Local-Net and Context-Net, respectively, to extract features. Both Local-Net and Context-Net consist of L shared frozen Transformer layers pretrained from the vision foundation model. Each layer of Local-Net is equipped with an MRLA, while each layer of Context-Net is integrated with a MRCA, enabling adaptation to objects at different scales within the local and context patches, respectively. After feature extraction, contextual cues are injected into the local representation through an Injection-Net, followed by producing local segmentation results. Finally, a CSAS leverages the sensitivity of Context-Net to large-scale semantic regions to guide Local-Net in focusing on these relevant areas.

patterns from multiple receptive fields to achieve deeper comprehension, due to the large spatial extent and rich structural details inherent in such images. To enhance image segmentation performance, adapters should process visual features across diverse spatial scales. To this end, we design an MRLA for local image patches, which mainly consists of multiple depthwise convolutions with varying kernel sizes and a 1×1 convolution. This design enables the model to capture multiscale spatial information with minimal trainable parameter overhead. In particular, the local patch P is fed into Local-Net to extract local features. The output of the attention operation or FFN in the ith layer is denoted as $\mathbf{X}_P^i \in \mathbb{R}^{d_i \times h_i \times w_i}$, which is passed to the MRLA. Within MRLA, $\mathbf{X}_P^i$ is first downprojected to reduce trainable parameters and computational complexity

$$\dot{\mathbf{X}}_P^i = \mathbf{W}_{Pdown}^i \mathbf{X}_P^i \tag{1}$$

where $\mathbf{W}_{Pdown}^i \in \mathbb{R}^{d_P \times d_i}$ ($d_P \ll d_i$) is the learnable parameters of the downprojection layer. Subsequently, three depthwise convolutions with different receptive fields (3×3, 5×5, and 7×7) are applied to the downprojected feature $\dot{\mathbf{X}}_P^i$. Their outputs are averaged and then passed through a 1×1 convolution to aggregate features across channels. Skip connections are incorporated around these operations. This process can be formulated as

$$\dot{\mathbf{X}}_P^i = \dot{\mathbf{X}}_P^i + \text{Avg}\left(\sum_{k=1}^{3} \text{DWConv}_{\omega_k \times \omega_k}\left(\dot{\mathbf{X}}_P^i\right)\right)$$

$$\dot{\mathbf{X}}_P^i = \dot{\mathbf{X}}_P^i + \text{Conv}_{1\times 1}\left(\dot{\mathbf{X}}_P^i\right) \tag{2}$$

where $\text{Avg}(\cdot)$ is an average operation, $\text{DWConv}_{\omega_k \times \omega_k}(\cdot)$ denotes a depthwise convolution with $\omega_k \times \omega_k$ kernel size and $\text{Conv}_{1\times 1}(\cdot)$ refers to a 1×1 convolution. Compared to standard convolutions, depthwise convolutions significantly reduce the number of parameters. Finally, the feature is up-projected back to the original dimension, and the original feature is added via a skip connection

$$\ddot{\mathbf{X}}_P^i = \mathbf{W}_{Pup}^i \dot{\mathbf{X}}_P^i + \mathbf{X}_P^i \tag{3}$$

where $\mathbf{W}_{Pup}^i \in \mathbb{R}^{d_i \times d_P}$ is the learnable parameters of the up-projection layer. As shown in Fig. 2, after adaptation by MRLA, Local-Net captures fine-grained spatial information.

C. Multireceptive Context Adapter

We first downsample the context patch to maintain the same spatial resolution as the local patch, thereby reducing computational cost. Compared to the local patch, the context patch has a broader field of view, which can provide valuable semantic guidance for the local patch. However, this inevitably introduces redundant information, which may interfere with or even mislead the representation of local features [11]. To address this issue, we propose an MRCA, which employs multiple depthwise dilated convolutions with different dilation rates. This design captures contextual features at varying scales and avoids excessive response to similar or repetitive regions,

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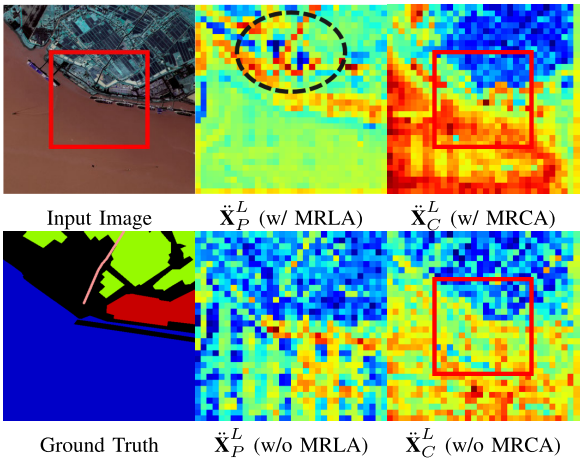


Fig. 2. Illustration of representative feature maps from the Lth layer with and without MRLA and MRCA. $\ddot{\mathbf{X}}_P^L$ with MRLA from Local-Net effectively captures local details (dashed black ellipse), while $\ddot{\mathbf{X}}_C^L$ with MRCA from Context-Net focuses on large-scale regions. The red box denotes the region in the context patch that corresponds to the local patch.

thereby more effectively extracting key contextual cues. In particular, Context-Net processes the context patch C to derive contextual representations. At the ith layer, the features entering the MRCA are represented as $\mathbf{X}_C^i \in \mathbb{R}^{d_i \times h_i \times w_i}$. Similar to MRLA, the MRCA commences with a downprojection layer to reduce the channel dimension

$$\dot{\mathbf{X}}_C^i = \mathbf{W}_{Cdown}^i \mathbf{X}_C^i \tag{4}$$

where $\mathbf{W}_{Cdown}^i \in \mathbb{R}^{d_C \times d_i}$ ($d_C \ll d_i$) denotes the learnable parameters of the downprojection layer. This is followed by three dilated depthwise convolutions with dilation rates of 1, 2, and 3, respectively. These features are fused via elementwise average and then processed by a 1×1 convolution for channelwise aggregation. A skip connection is applied to preserve the identity. This process can be expressed as follows:

$$\begin{aligned}\dot{\mathbf{X}}_C^i &= \dot{\mathbf{X}}_C^i + \text{Avg}\left(\sum_{k=1}^{3} \text{DWConv}_{3\times3}\left(\dot{\mathbf{X}}_C^i; r_k\right)\right) \\ \dot{\mathbf{X}}_C^i &= \dot{\mathbf{X}}_C^i + \text{Conv}_{1\times1}\left(\dot{\mathbf{X}}_C^i\right)\end{aligned} \tag{5}$$

where $\text{DWConv}_{3\times3}(\cdot; r_k)$ denotes a 3×3 dilated depthwise convolution with dilation rate r_k. Dilated depthwise convolutions can skip redundant features and focus on key cues by sparse sampling. Finally, the feature is projected back to the original dimension and added to the original feature via a skip connection

$$\ddot{\mathbf{X}}_C^i = \mathbf{W}_{Cup}^i \dot{\mathbf{X}}_C^i + \mathbf{X}_C^i \tag{6}$$

where $\mathbf{W}_{Cup}^i \in \mathbb{R}^{d_i \times d_C}$ is the learnable parameters of the upprojection layer. As shown in Fig. 2, equipped with MRCA, Context-Net provides comprehensive contextual cues.

D. Injection-Net

After processing through the Lth layer of Local-Net and Context-Net, local features $\ddot{\mathbf{X}}_P^L$ and contextual features $\ddot{\mathbf{X}}_C^L$ are extracted, respectively. To effectively fuse local spatial details with contextual semantics, Injection-Net is responsible for injecting contextual cues into the local feature representations. This injection process preserves detailed spatial structures while enhancing the semantic perception of local features over broader regions, ultimately generating cooperative features that integrate both local details and contextual awareness.

In practice, we employ deformable attention [57] to fuse contextual information into local features

$$\begin{aligned}\ddot{\mathbf{X}}_{PC}^L &= \text{DeformAttn}\left(\mathbf{Q}, \mathbf{KV}\right) \\ &= \text{DeformAttn}\left(\ddot{\mathbf{X}}_P^L, \ddot{\mathbf{X}}_C^L, \ddot{\mathbf{X}}_C^L\right) \\ \ddot{\mathbf{X}}_{PC}^L &= \gamma \ddot{\mathbf{X}}_{PC}^L + \ddot{\mathbf{X}}_P^L\end{aligned} \tag{7}$$

where γ is a learnable variable initialized to zero. This efficient attention mechanism reduces computational cost through sparse sampling while dynamically selecting the most critical contextual cues. Finally, the cooperative features $\ddot{\mathbf{X}}_{PC}^L$ and multiscale shallow features are fed into the decoder to generate the local prediction result.

E. Context-Sensitive Assistance Strategy

Although the proposed framework effectively leverages both local and contextual information, existing optimization strategies do not consider the differential contributions of different local regions to semantic discrimination. This naturally reduces the models sensitivity to large-scale regions such as water bodies and forests.

To address this issue, we introduce a CSAS. In particular, the multiscale features from the Context-Net are fed into a new decoder to generate the context result during training. We first crop the central part of the context result $\mathbf{R}_C$ that is spatially aligned with the local result $\mathbf{R}_P$, denoted as $\hat{\mathbf{R}}_C$. The context patch has a broader field of view, and thus the focus of the cropped result $\hat{\mathbf{R}}_C$ is on large-scale semantic regions, which can assist the Local-Net in enhancing its sensitivity to such areas. Then, we convert it into a pseudolabel

$$\hat{\mathbf{Y}}_C = \text{Up}\left(\text{Argmax}\left(\hat{\mathbf{R}}_C\right)\right) \tag{8}$$

where $\text{Argmax}(\cdot)$ is performed along the channel dimension and $\text{Up}(\cdot)$ is an upsampling operation. As shown in Fig. 1, the pseudolabel $\hat{\mathbf{Y}}_C$ and the local ground truth semantic map $\mathbf{Y}_P$ are fed simultaneously into the mask selection module. We adopt the XNOR operation to identify the positions where $\hat{\mathbf{Y}}_C$ matches $\mathbf{Y}_P$

$$\mathbf{M} = \text{XNOR}\left(\hat{\mathbf{Y}}_C, \mathbf{Y}_P\right) \tag{9}$$

where a value of 1 in the mask $\mathbf{M}$ indicates a correct prediction by Context-Net. Subsequently, we extract the elements from $\mathbf{Y}_P$ and $\mathbf{R}_P$ based on $\mathbf{M}$

$$\begin{aligned}\Omega_P &= \left\{(i,j) \mid 1 \le i \le H_P,\ 1 \le j \le W_P\right\} \\ \mathbf{R}_P^M &= \left\{(\mathbf{R}_P)_{ij} \mid (i,j) \in \Omega_P,\ \mathbf{M}_{ij} = 1\right\} \\ \mathbf{Y}_P^M &= \left\{(\mathbf{Y}_P)_{ij} \mid (i,j) \in \Omega_P,\ \mathbf{M}_{ij} = 1\right\}.\end{aligned} \tag{10}$$

Finally, the selected prediction-label pairs are used to compute an additional supervision loss for Local-Net

$$\mathcal{L}_{mce} = \mathcal{L}_{ce}\left(\mathbf{R}_P^M, \mathbf{Y}_P^M\right) \tag{11}$$

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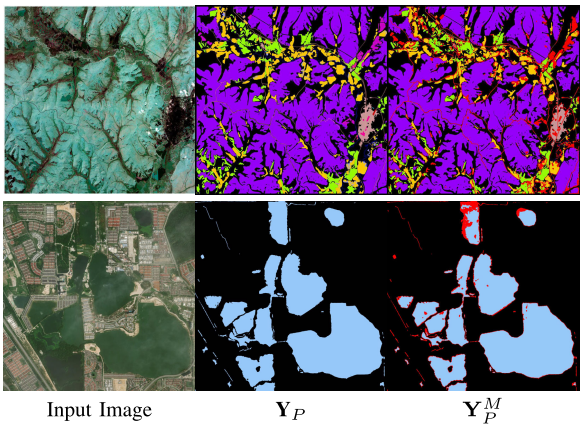


Fig. 3. Training-time visualization of the selected ground truth $\mathbf{Y}_P^M$, where red represents the excluded regions. Compared to the ground truth $\mathbf{Y}_P$, $\mathbf{Y}_P^M$ emphasizes large-scale semantic regions. Please zoomed-in view for a better view.

where $\mathcal{L}_{ce}(\cdot,\cdot)$ denotes the cross-entropy loss. This loss leverages the distinct regions of interest for Local-Net and Context-Net, effectively overcoming the primary limitations of the patch-based training paradigm. Under large receptive fields, the correct predictions from Context-Net reflect the semantic understanding of broad regions. As shown in Fig. 3, compared to $\mathbf{Y}_P$, the selected ground truth $\mathbf{Y}_P^M$ mainly focuses on large-scale semantic regions. This enhances Local-Net's sensitivity to context-dependent regions.

In addition, Local-Net and Context-Net are supervised using cross-entropy loss

$$\begin{aligned}\mathcal{L}_{pce} &= \mathcal{L}_{ce}\big(\mathbf{R}_P, \mathbf{Y}_P\big)\\ \mathcal{L}_{cce} &= \mathcal{L}_{ce}\big(\mathbf{R}_C, \mathbf{Y}_C\big)\end{aligned} \tag{12}$$

where $\mathbf{Y}_C$ is the context ground truth semantic map. Hence, the overall objective can be expressed as follows:

$$\mathcal{L} = \mathcal{L}_{pce} + \mathcal{L}_{cce} + \beta\mathcal{L}_{mce} \tag{13}$$

where β is a hyperparameter used to balance the loss.

IV. Experimental Dataset and Setup

In this section, we first introduce the three publicly available UHR remote sensing image datasets in our experiments, including five-billion-pixels (FBPs) [3], GLH-Water [5], and Inria Aerial [1], which are used to evaluate the effectiveness of the parameter-efficient local-context cooperation (PEACE) framework. Fig. 4 presents example images and corresponding ground truth selected from these datasets, while the pixelwise class distribution across the datasets is summarized in Fig. 5. Subsequently, we provide a detailed description of the datasets, experimental setup, and evaluation metrics.

A. Datasets

1) FBP Dataset: The FBP dataset [3] is a challenging benchmark designed for large-scale land cover mapping, with severe class imbalance [as shown in Fig. 5(a)]. It provides 150 UHR Gaofen-2 satellite images, each with a spatial resolution of 6800 × 7200 pixels. The dataset consists of

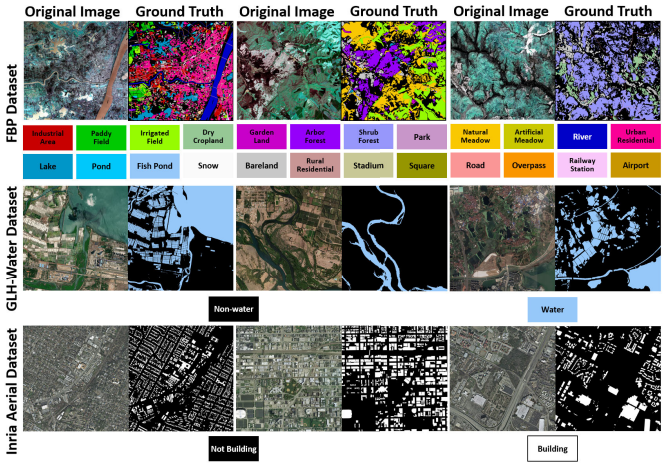


Fig. 4. Example images and corresponding ground truth from the FBP [3], GLH-Water [5], and Inria Aerial [1] datasets.

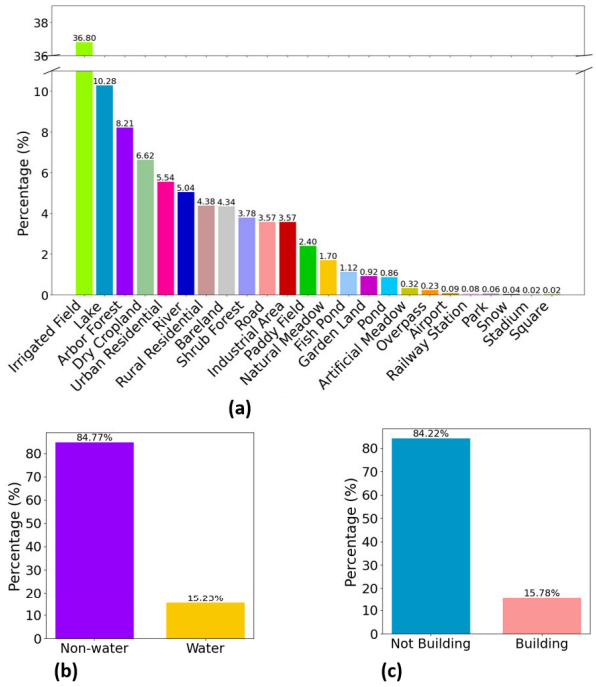


Fig. 5. Pixelwise class distribution in (a) FBP [3] dataset, (b) GLH-Water [5] dataset, and (c) Inria Aerial [1] dataset.

24 annotated categories, covering artificial-constructed, agricultural, and natural classes. Following the official data split [3], we use 120 images for training and 30 images for testing.

2) GLH-Water Dataset: The GLH-Water dataset [5] is designed for global surface water detection in large-scale UHR satellite imagery, facilitating research and applications such as refined flood mapping and comprehensive water resource assessment. It consists of 250 UHR satellite images with a massive size of 12 800 × 12 800 pixels. The dataset focuses on binary segmentation, labeling pixels as either water or nonwater. We follow the official data split [5], i.e., the number of training images, validation images, and testing images is 200, 25, and 25, respectively.

3) Inria Aerial Dataset: The Inria Aerial dataset [1] covers diverse urban landscapes from five cities for building detection, ranging from dense metropolitan districts to alpine

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TABLE I

COMPARISON RESULTS ON FBPS [3], GLH-WATER [5], AND INRIA AERIAL [1]. PARAM* REPRESENTS THE TRAINABLE PARAMETERS IN THE BACKBONE. THE VALUES IN BOLD INDICATE THE BEST RESULTS

Generic Models	Backbone	Param* (M)	FBP		GLH-Water		Inria Aerial	
			mIoU (%)	mF1 (%)	IoU (%)	F1 (%)	IoU (%)	F1 (%)
DeepLabv3+ [32]	ResNet50 [60]	23.5	67.0	78.8	79.0	88.2	81.9	90.0
SegFormer [37]	MiT-B5 [37]	81.4	69.4	80.7	84.1	91.4	82.5	90.4
UPerNet [59]	Swin-Base [14]	86.9	69.2	80.4	84.2	91.4	82.6	90.5
MaSSFormer [39]	Swin-Base [14]	86.9	68.3	79.9	84.0	91.3	82.3	90.3
Ultra-high Resolution Models								
GLNet [10]	ResNet50 [60]	-	-	-	-	-	71.2	82.4
FCtL [25]	VGG16 [61]	41.5	44.5	60.1	74.9	85.7	73.7	84.1
ISDNet [7]	ResNet18 [60]+STDC1 [40]	13.8	62.7	77.6	82.1	90.2	74.2	84.9
WSDNet [8]	ResNet18 [60]+STDC1 [40]	-	-	-	-	-	75.2	86.0
GPWFormer [20]	ResNet18 [60]+Wave-ViT [62]	-	-	-	-	-	76.5	86.2
BPT [31]	PMT [31]	-	-	-	-	-	77.1	86.3
SGHRQ [21]	ResNet18 [60]+STDC1 [40]	13.8	59.3	71.3	82.2	90.2	77.3	87.2
FCtL++ [11]	MiT-B2 [37]	-	-	-	-	-	78.1	87.7
PCL [5]	ResNet50 [60]	-	-	-	82.2	-	-	-
RUE [9]	ResNet18 [60]+STDC1 [40]	13.8	63.5	75.4	73.2	84.5	79.0	88.0
HLRC [22]	Swin-Tiny [14]+EfficientViT [42]	32.5	68.6	79.4	86.3	92.7	80.8	89.4
Ours	Swin-Base [14] (Swin Transformer [14])	9.4	70.3	81.0	87.9	93.6	82.0	90.1
Ours	Hiera-Base+ [58] (SAM2 [15])	8.2	70.2	81.1	87.5	93.3	**82.8**	**90.6**
Ours	ViT-Base [36] (DINOv2 [16])	**6.5**	72.1	82.5	86.9	93.0	82.2	90.2
Ours	ViT-Base [36] (DINOv3 [17])	**6.5**	**72.4**	**82.9**	**89.3**	**94.4**	82.6	90.5

resorts. It consists of 180 UHR aerial images of 5000 × 5000 pixels with the binary mask for building/nonbuilding areas. Following the protocol of [25], we split images into training, validation, and testing sets with 126, 27, and 27 images, respectively.

B. Experimental Settings

We evaluate our framework on a single NVIDIA A40 GPU. All experiments are conducted in Python 3.11.10, CUDA 12.2, and PyTorch 2.0.1. We adopt three distinct vision foundation models to verify the universality, including Swin Transformer [14] (pretrained on image classification), SAM2 [15] (pretrained on a large-scale video segmentation dataset), and both DINOv2 [16] and DINOv3 [17] (pretrained via self-supervised learning on curated datasets). In particular, we employ the Swin-Base [14], Hiera-Base+ [58], and ViT-Base [36] architectures for Swin Transformer, SAM2, and the DINO series as our backbone networks, respectively. The UPerNet [59] is used as the segmentation decoder. During training, we use the AdamW optimizer with an initial learning rate of 0.0003 and a weight decay of 0.01. The learning rate is scheduled using a polynomial decay policy with a power of 1.0, and the batch size is set to 8. All models are trained for 80K iterations with 1.5K iterations of linear warm-up. Data augmentation techniques include random cropping, random flipping, and photometric distortion. For inference, the UHR remote sensing images are divided into local patches of 512 × 512 pixels, with a 25% overlap between adjacent patches. The spatial resolution of context patches is 1024 × 1024 pixels.

C. Evaluate Metrics

We adopt intersection over union (IoU) and $F1$-score ($F1$) to evaluate our framework, which are defined as follows:

$$\begin{aligned} \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \\ \text{Precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{F}_1 &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}} \end{aligned} \tag{14}$$

where TP, TN, FP, and FN denote the number of true positive pixels, true negative pixels, false positive pixels, and false negative pixels, respectively. We compute mean IoU (mIoU) and mean F1 (mF1) across all classes on the FBP dataset. For the GLH-Water dataset, we report IoU and $F1$ for the water class only. Similarly, for the Inria Aerial dataset, we report IoU and $F1$ for the building class only.

V. EXPERIMENTAL RESULT

In this section, we first compare with previous state-of-the-art methods, covering both generic semantic segmentation models and UHR semantic segmentation methods. In addition, we evaluate the generalization capability of our framework on surface water detection using RSFMs. Finally, we conduct ablation experiments to demonstrate the effectiveness of the main components and explore model efficiency.

A. Comparison With State-of-the-Art Methods

For evaluation, we compare our framework against DeepLabv3 + [32] with ResNet50 [60], SegFormer [37] with MiT-B5 [37], UPerNet [59] with Swin-Base [14], MaSSFormer [39] with Swin-Base [14], GLNet [10], FCtL [25], ISDNet [7], WSDNet [8], GPWFormer [20], BPT [31], SGHRQ [21], FCtL++ [11], PCL [5], RUE [9], and HLRC [22]. Among these methods, DeepLabv3+, SegFormer, UPerNet, and MaSSFormer are generic models for semantic segmentation, while the rest are tailored for UHR remote sensing image segmentation and are thus denoted as UHR

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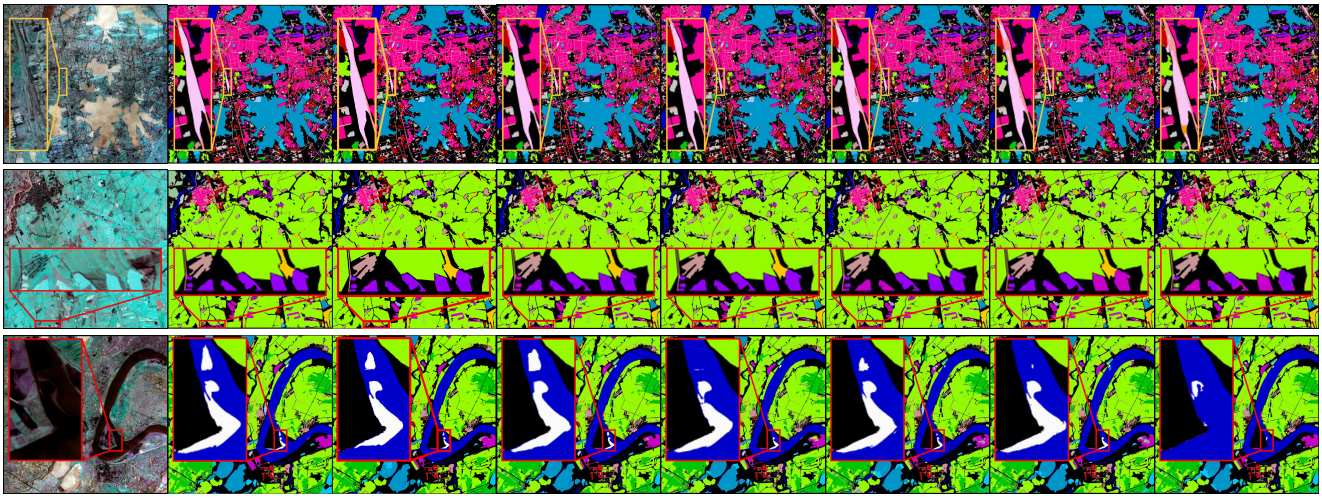


Fig. 6. Comparison of our results with SegFormer [37] and HLRC [22] on the FBP dataset [3]. On (first row), light pink represents "railway station" and orange pink represents "Road". On (second row), dark purple represents "arbor forest", and dark pink represents "garden land". On (last row), white represents "snow".

models. In particular, we retrain and test all generic models on three datasets to establish baseline results, employing patch-based inference to balance performance and computational resource requirements. For the UHR models, we provide the reported results from the respective papers on the Inria Aerial dataset. On the FBP and GLH-Water datasets, we retrain and test FCtL, ISDNet, SGHRQ, RUE, and HLRC for comparison. In addition, we follow the reported results from PCL on the GLH-Water dataset.

1) Results on the FBP Dataset: The FBP dataset [3] is annotated with a wide range of semantic categories and contains objects at various scales, posing significant challenges for UHR remote sensing image segmentation. Therefore, we first conduct experiments on this dataset, with results summarized in Table I. Generic models achieve generally strong performance, among which SegFormer [37] performs best with an mIoU of 69.4% and an mF1 of 80.7%. However, it requires 81.4M trainable parameters in the backbone. On the contrary, most UHR models underperform, with only HLRC [22] achieving competitive results (i.e., 68.6% mIoU and 79.4% mF1), yet still involving 32.5M trainable backbone parameters. Our proposed framework effectively integrates local and contextual cues and achieves superior performance, surpassing all previous methods. Notably, the ViT backbone pretrained via DINOv3 [17] exhibits superior interclass discriminative capacity. Equipped with these features, our approach achieves an improvement of +3.0% in mIoU and +2.2% in mF1 over SegFormer. In addition, by adopting the parameter-efficient fine-tuning paradigm, the number of trainable parameters in our backbone remains below 10M, reducing to only 6.5M when using the DINO series.

To analyze the ability to identify challenging classes, we list the IoU of representative categories divided into three groups in Table II. The overpass and railway station are often misclassified as the road due to structural connectivity and spatial overlap. Our DINO-based variants achieve the best overall results, with DINOv2 obtaining the highest mIoU of 70.6%. The natural meadow, artificial meadow, and arbor forest are frequently confused because of similar vegetation characteristics. All of our results achieve higher overall accuracy than previous methods, demonstrating improved discrimination of subtle botanical differences. The snow, stadium, and square are the classes with the rarest samples in the dataset. Compared with the previous state-of-the-art approach, SegFormer, our DINOv3-based method achieves a 1.3% higher mIoU. This indicates that the rich pretrained features from DINOv3 effectively alleviate the issue of data scarcity. As illustrated in Fig. 6, we present several qualitative comparison

TABLE II

RESULTS FOR CHALLENGING CLASSES ON THE FBP DATASET [3]. THE ABBREVIATIONS FOR CATEGORIES ARE DEFINED AS: OVER. - OVERPASS, RAIL. - RAILWAY STATION, NATU. - NATURAL MEADOW, ARTI. - ARTIFICIAL MEADOW, ARBO. - ARBOR FOREST, STAD. - STADIUM, SQUA. - SQUARE. THE VALUES IN BOLD INDICATE THE BEST RESULTS

Model	Over.	Road	Rail.	mIoU(%)
SegFormer [37]	73.7	75.7	55.4	68.3
HLRC [22]	71.1	75.0	52.8	66.3
Ours (Swin [14])	73.5	76.3	56.8	68.9
Ours (SAM2 [15])	74.0	**76.5**	54.2	68.2
Ours (DINOv2 [16])	74.5	76.1	**61.1**	**70.6**
Ours (DINOv3 [17])	**75.0**	76.2	59.1	70.1
Model	Natu.	Arti.	Arbo.	mIoU(%)
SegFormer [37]	88.3	59.2	**96.1**	81.2
HLRC [22]	87.9	62.6	95.3	81.9
Ours (Swin [14])	88.0	62.2	95.8	82.0
Ours (SAM2 [15])	88.9	**63.1**	95.2	82.4
Ours (DINOv2 [16])	89.7	62.3	96.0	82.6
Ours (DINOv3 [17])	**90.9**	62.9	**96.1**	**83.3**
Model	Snow	Stad.	Squa.	mIoU(%)
SegFormer [37]	63.0	59.4	**37.7**	53.4
HLRC [22]	23.6	52.7	24.9	33.7
Ours (Swin [14])	60.2	52.4	32.8	48.5
Ours (SAM2 [15])	63.4	56.7	29.8	50.0
Ours (DINOv2 [16])	**74.3**	54.4	35.2	54.6
Ours (DINOv3 [17])	65.9	**60.6**	37.6	**54.7**

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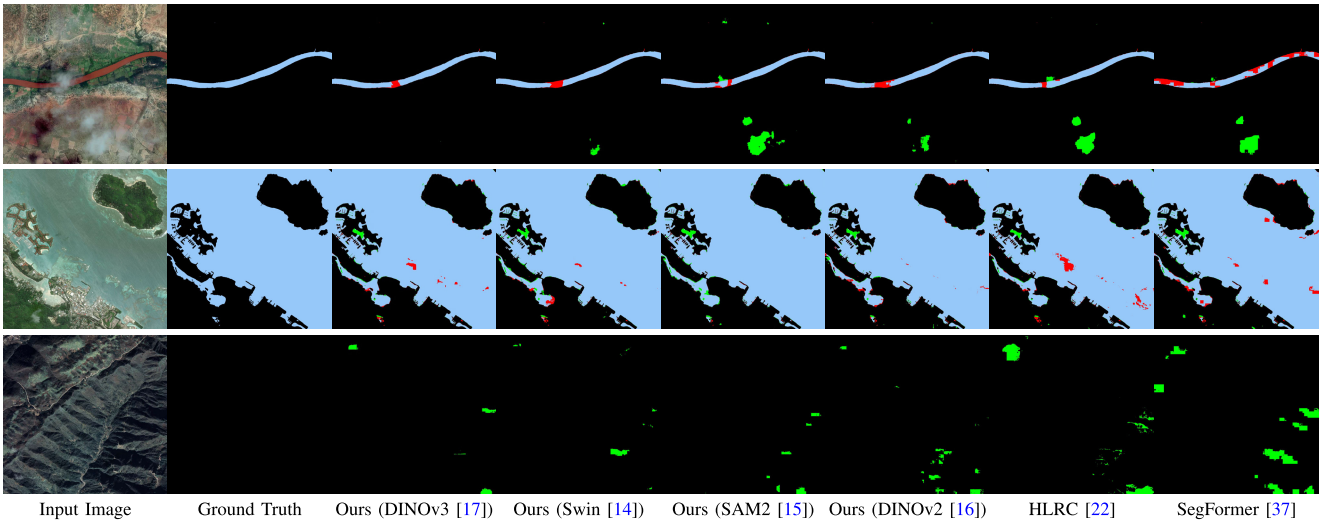


Fig. 7. Comparison of our results with HLRC [22] and SegFormer [37] on the GLH-water dataset [5]. The black regions refer to the background, while the light blue regions represent the water bodies. Besides, the red and green pixels represent false negatives and false positives, respectively.

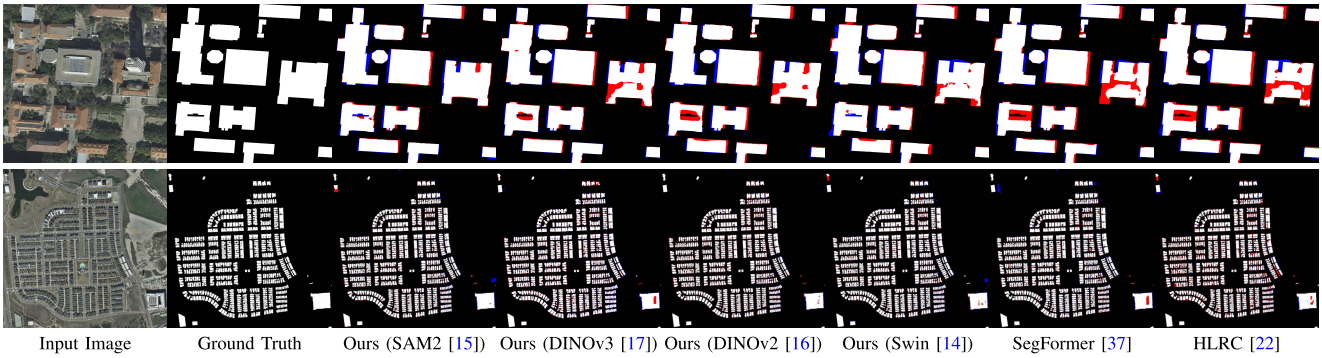


Fig. 8. Comparison of our results with SegFormer [37] and HLRC [22] on the Inria Aerial dataset [1]. The black regions refer to the background, while the white regions represent the extracted buildings. Besides, the red and blue pixels represent false negatives and false positives, respectively.

results. The results demonstrate that our method achieves clearer discrimination between similar categories and more accurate recognition for classes with rare samples.

2) Results on the GLH-Water Dataset: The GLH-Water dataset [5] is characterized by its extremely high spatial resolution, with individual images reaching up to 12K $\times$ 12K pixels. However, the extensive coverage of water bodies often leads to difficulties in capturing sufficient contextual information. This presents substantial challenges for the identification of large-scale water bodies. Our proposed framework introduces critical contextual information, thus accurately capturing the spatial continuity of water bodies while maintaining their integrity. As shown in Table I, our results demonstrate significant advantages in handling these large-scale water regions compared to existing methods. In particular, our DINOv3-based model achieves the best performance, outperforming generic models by at least 5.1% in IoU and 3.0% in $F1$. Furthermore, compared against UHR models, our DINOv3-based approach surpasses the previous state-of-the-art model, HLRC [22], by 3.0% in IoU and 1.7% in $F1$. We present several qualitative comparison results in Fig. 7. It can be observed that our method performs better in capturing slender water bodies, large water surfaces, and nonwater areas. Moreover, even under cloud cover, our method demonstrates superior capability in reasoning the presence of water.

3) Results on the Inria Aerial Dataset: The Inria Aerial dataset [1] features a dense distribution of small buildings while containing only a few large-scale structures. Since the required contextual information is relatively limited, generic models already perform well on this dataset, leaving limited headroom for improvement in overall IoU. As shown in Table I, UPerNet [59] with Swin-Base [14] achieves competitive results and attains 82.6% IoU and 90.5% $F1$. On the contrary, UHR models often underperform due to strategies such as downsampling that lead to loss of boundary precision. Our proposed framework emphasizes local patterns and fine-grained structures, enabling more precise boundary delineation of small objects. In particular, our SAM2-based approach achieves the highest performance and reaches 82.8% IoU and 90.6% $F1$ due to the powerful pixel-level mask generation capability of SAM2 [15]. In Fig. 8, we illustrate two examples comparing our approach with SegFormer [37] and HLRC [22]. The results show that our model more accurately captures edge contours, especially the SAM2-based model.

4) Comparison With REST Using RSFM: REST [30] is a UHR segmentation method and achieves its best performance

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TABLE III

PERFORMANCE COMPARISON WITH REST [30] USING RSFM. IOU (%) IS REPORTED AND THE BEST RESULTS ARE IN BOLD

Model	RSFM	GLH-Water	FBP
REST [30]	SkySense [18]	89.15	**72.95**
Ours	SkySense [18]	**90.46**	72.84

TABLE IV

CROSS-DATASET GENERALIZATION FOR SURFACE WATER DETECTION USING RSFM. NOTE THAT MODEL IS TRAINED ONLY ON GLH-WATER [5] AND TESTED ON GLH-WATER, FBP [3], AND LOVEDA [63], RESPECTIVELY. FOR FBP AND LOVEDA, THE PREDICTED WATER ID IS MAPPED TO THE CORRESPONDING WATER-CLASS ID FOR EVALUATION. IOU (%) IS REPORTED AND THE BEST RESULTS ARE IN BOLD

Model	RSFM	Train	Test		
			GLH-Water	FBP	LoveDA
REST [30]	SkySense [18]	GLH-Water	89.15	47.69	75.56
Ours	SkySense [18]	GLH-Water	90.46	69.71	80.52
Ours	SkySense++ [19]	GLH-Water	**90.54**	**71.07**	**81.40**

TABLE V

COMPARISON OF GPU MEMORY USAGE, TOTAL PARAMETERS (PARAM), AND FLOPS. GPU MEMORY USAGE IS MEASURED DURING TESTING WITH A BATCH SIZE OF 1, AND "–" INDICATES MULTI-GPU INFERENCE. THE FLOPS ARE CALCULATED ON 512×512 INPUT PATCHES

Model	Memory(MB)	Param(M)	FLOPs(G)
SegFormer [37]	1111	82.0	51.9
HLRC [22]	2840	66.1	28.1
REST [30]	-	271.9	475.4
Ours	1777	125.3	398.4

using SkySense [18], a domain-specific RSFM. Therefore, we compare our framework with REST under the same RSFM setting to ensure fairness. As shown in Table III, our method outperforms REST by 1.31% mIoU on GLH-Water [5] and achieves comparable performance on FBP [3]. This highlights the competitiveness of our framework in maintaining semantic coherence.

In addition, we perform cross-dataset evaluation under the same RSFM setting to validate model generalization, where REST is tested with the official weights using two A40 GPUs. In particular, we analyze cross-dataset generalization by training only on GLH-Water and testing on GLH-Water, FBP, and LoveDA [63], respectively. During evaluation on FBP and LoveDA, the model trained on GLH-Water produces binary water/nonwater predictions. The predicted water ID is then mapped to the corresponding water-class ID in each dataset. IoU and $F1$ are computed for the water class based on the original annotations. As shown in Table IV, our method achieves pronounced gains on FBP and a dramatic improvement on LoveDA, demonstrating superior cross-dataset generalization capability. Moreover, using a more advanced RSFM (SkySense++ [19]) yields consistent improvements, indicating that a stronger RSFM can further enhance cross-dataset generalization.

5) Inference Efficiency: Table V reports the inference efficiency comparison in terms of GPU memory usage, total parameters, and FLOPs. SegFormer [37], as a representative generic model, has advantages in efficiency. Although our method is not superior in terms of total parameters or FLOPs, its test-time GPU memory remains below 2 GB, which is still acceptable for low-memory devices. Among UHR methods, HLRC [22] and REST [30] are competitive methods. Compared with HLRC, our method requires less GPU memory during testing. Compared with REST, our method is more efficient in both total parameters and FLOPs, and does not require multi-GPU inference. Combined with Tables I and III, these comparisons suggest that our method achieves a favorable tradeoff between performance and efficiency.

TABLE VI

COMPARISON OF INFERENCE TIME ON A SINGLE IMAGE WITH A RESOLUTION OF 6800 × 7200 PIXELS

Model	Overlapping	mIoU(%)	Time(s)
SegFormer [37]	√	69.4	13.7
SegFormer [37]	×	$68.0_{\downarrow 1.4}$	8.4
Ours (DINOv2 [16])	√	72.1	23.9
Ours (DINOv2 [16])	×	$71.3_{\downarrow 0.8}$	14.7

TABLE VII

ABLATION STUDY ON OUR PEACE FRAMEWORK. PARAM* REPRESENTS THE TRAINABLE PARAMETERS IN THE BACKBONE. THE FLOPS ARE CALCULATED ON 512×512 INPUT PATCHES. THE VALUES IN BOLD INDICATE THE BEST RESULTS. MRLA: MULTIRECEPTIVE LOCAL ADAPTER; MRCA: MULTIRECEPTIVE CONTEXT ADAPTER; CSAS: CONTEXT-SENSITIVE ASSISTANCE STRATEGY

Model	Param*(M)	FLOPs(G)	mIoU(%)	mF1(%)
Baseline	86.6	305.1	68.9	80.2
+ Freeze Backbone	0.0	305.1	$66.4_{-2.5}$	$78.2_{-2.0}$
+ MRLA	2.7	307.8	$69.2_{+0.3}$	$80.4_{+0.2}$
+ Context Input	6.6	398.5	$70.8_{+1.9}$	$81.4_{+1.2}$
+ MRCA	6.5	398.4	$71.4_{+2.5}$	$82.1_{+1.9}$
+ CSAS	6.5	398.4	$\mathbf{72.1}_{+3.2}$	$\mathbf{82.5}_{+2.3}$

We further compare the inference time with SegFormer on a single image of resolution 6800 × 7200 pixels. As shown in Table VI, our total inference time is longer than that of SegFormer. To address this, we partition the image into nonoverlapping local patches during testing. With this strategy, the mIoU decreases by only 0.8%, yet it remains 1.9% higher than SegFormer, while the total inference time becomes comparable to SegFormer. Similarly, we test SegFormer with nonoverlapping local patches. Although this approach further improves its inference speed, it leads to a 1.4% drop in mIoU. This result demonstrates that incorporating contextual information effectively mitigates the boundary issues in local patch processing.

B. Ablation Experiments

We perform ablation studies on the FBP dataset [3] to assess the effectiveness of each component and analyze model efficiency.

1) Ablation for Our Overall Framework: We adopt ViT-Base [36] pretrained with DINOv2 [16] as the backbone of the Local-Net for processing local patches, and employ UPerNet [59] as the decoder in our experiments. As shown

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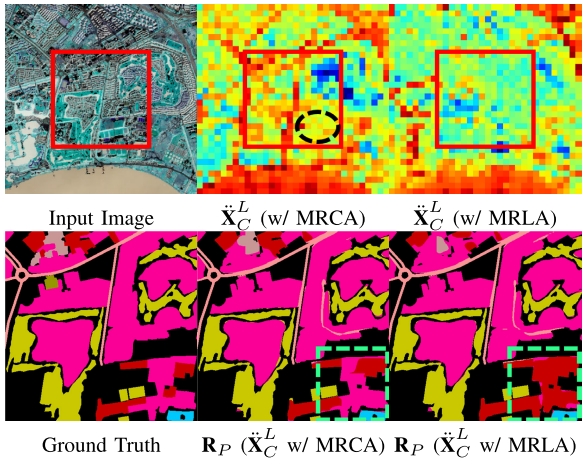


Fig. 9. Illustration of representative feature maps from the Lth layer of Context-Net with MRCA and MRLA. $\ddot{\mathbf{X}}_C^L$ with MRCA produces stronger responses at semantically ambiguous regions (dashed black ellipse). The red box denotes the region in the context patch that corresponds to the local patch.

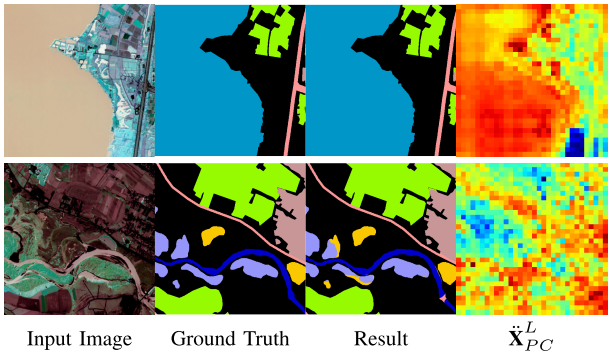


Fig. 10. Visualization of cooperative features $\ddot{\mathbf{X}}_{PC}^L$ that integrate both local and contextual information.

in Table VII, as a reference, we first construct a baseline by fine-tuning all 86.6M parameters of the Local-Net, achieving a performance of 68.9% mIoU and 80.2% mF1. Building upon this baseline, we then freeze the backbone to minimize the number of trainable parameters in Local-Net. However, this causes mIoU and mF1 to drop by 2.5% and 2.0% due to the lack of task-specific optimization.

To address this, we insert the MRLA into the Local-Net to enable task-specific adaptation with minimal trainable parameters and effectively capture local patterns. Compared to the baseline, our approach improves performance to 69.2% mIoU and 80.4% mF1. Next, we introduce a Context-Net equipped with MRLA to process context patches and pass context information through an Injection-Net, aiming to alleviate the limited receptive field inherent in local patch processing. This results in noticeable improvements of 1.9% in mIoU and 1.2% in mF1 compared to the baseline. However, as contextual regions may contain redundant information that interferes with the semantic understanding of local patches [11], we replace the MRLA in Context-Net with the proposed MRCA. MRCA is designed to extract critical contextual cues. As shown in Fig. 9, we visualize the representative feature maps from Context-Net with MRCA and MRLA. By replacing MRLA with MRCA, we observe stronger responses at semantically ambiguous regions (highlighted by the dashed black ellipse in the image in the first row and second column). This indicates that MRCA is more capable of emphasizing informative features, whereas MRLA is more susceptible to interference from erroneous and redundant semantic information, causing important cues to be diluted. This leads to significant performance gains, improving mIoU and mF1 by 2.5% and 1.9%, respectively, over the baseline, with negligible changes in GFLOPs and trainable parameters. In addition, we visualize the cooperative features in Fig. 10, demonstrating that they effectively capture both contextual cues and local details.

Finally, we incorporate the CSAS to enhance the sensitivity of Local-Net to large-scale regions. This results in further improvements of 3.2% in mIoU and 2.3% in mF1 compared to the baseline, achieving a final performance of 72.1% mIoU and 82.5% mF1. In summary, our approach incurs only a marginal overhead of 6.5M trainable parameters over the baseline model, yet achieves significant performance gains.

TABLE VIII

ABLATION STUDY ON PARAMETER-EFFICIENT FINE-TUNING STRATEGY. PARAM* REPRESENTS THE TRAINABLE PARAMETERS IN THE BACKBONE. THE VALUES IN BOLD INDICATE THE BEST RESULTS. MRLA: MULTIRECEPTIVE LOCAL ADAPTER; MRCA: MULTIRECEPTIVE CONTEXT ADAPTER

Local-Net	Context-Net	Param*(M)	mIoU(%)
/	/	1.2	67.2
LoRA [50]	MRCA	6.2	69.5
Adapter [49]	MRCA	6.2	69.6
MRLA	LoRA [50]	6.3	68.4
MRLA	Adapter [49]	6.3	69.2
MRLA	MRLA	6.6	71.2
MRCA	MRCA	6.4	69.3
MRLA	MRCA	6.5	**72.1**

2) Ablation for Parameter-Efficient Fine-Tuning Strategy: To validate the superiority of our proposed MRLA and MRCA, we conduct ablation studies by replacing each of them with other parameter-efficient fine-tuning strategies, including LoRA [50] and Adapter [49]. In particular, we replace MRLA with LoRA or Adapter, and separately replace MRCA with LoRA or Adapter. As shown in Table VIII, all results show a significant drop in mIoU, highlighting the importance of multiscale spatial information for both Local-Net and Context-Net. Furthermore, we conduct experiments by simultaneously adopting MRLA or MRCA in both Local-Net and Context-Net. The degradation in both scenarios indicates the differing roles of local patches and context patches in feature representation.

3) Ablation for Downprojection Dimension d_P and d_C: The downprojection dimensions d_P in the MRLA and d_C in the MRCA primarily determine the number of trainable parameters in the backbone. We conduct experiments to investigate the influence of d_P and d_C on model performance. First, we increase d_P and d_C simultaneously. As shown in Table IX, when the dimensions are increased to 64, the number of learnable backbone parameters rises and model accuracy improves consistently. However, further increasing the dimensions to 96 leads to a performance drop. Next, we perform ablation

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TABLE IX

ABLATION STUDY ON DOWNPROJECTION DIMENSION d_P AND d_C. PARAM* REPRESENTS THE TRAINABLE PARAMETERS IN THE BACKBONE. THE VALUES IN BOLD INDICATE THE BEST RESULTS

d_P	d_C	Param*(M)	mIoU(%)
32	32	3.9	68.5
48	48	5.2	69.8
64	64	6.5	**72.1**
96	96	9.2	70.5
64	32	5.2	69.9
64	48	5.9	71.0
64	64	6.5	**72.1**
64	96	7.9	70.0
32	64	5.2	68.3
48	64	5.9	70.3
64	64	6.5	**72.1**
96	64	7.9	70.3

TABLE X

ABLATION STUDY ON INJECTION METHOD. PARAM* REPRESENTS THE TRAINABLE PARAMETERS IN THE BACKBONE. THE VALUES IN BOLD INDICATE THE BEST RESULTS

Injection Method	Param*(M)	mIoU(%)
Addition	5.3	70.6
Concatenation	6.5	69.4
Cross Attention [36]	7.6	70.8
FreqFusion [64]	5.4	69.8
Deformable Attention [57]	6.5	**72.1**

TABLE XI

ABLATION STUDIES ON (A) SPATIAL RESOLUTION $H_C \times W_C$ OF THE CONTEXT PATCH AND (B) CONTEXT-NET BACKBONE. THE VALUES IN BOLD INDICATE THE BEST RESULTS

(a)

$H_C \times W_C$	/	768×768	1024×1024	1280×1280	1536×1536
mIoU (%)	69.2	70.4	**72.1**	70.8	70.0

(b)

Backbone (Context-Net)	ViT-Small	ViT-Base
mIoU (%)	69.6	**72.1**

TABLE XII

ABLATION STUDIES ON (A) SELECTION STRATEGIES OF CSAS AND (B) HYPERPARAMETER β. THE VALUES IN BOLD INDICATE THE BEST RESULTS

(a)

Method	/	$\hat{\mathbf{Y}}_C$	XOR	XNOR
mIoU (%)	71.4	69.3	70.7	**72.1**

(b)

β	/	0.5	1	2	5
mIoU (%)	71.4	71.5	71.9	**72.1**	70.9

studies by fixing one of d_P or d_C at 64 while varying the other. We observe similar trends, with performance saturating at dimension 64 and additional adapter parameters yielding no further gains. We argue that lower dimensions may suffer from insufficient representation, limiting the model's adaptability, whereas higher dimensions may dilute the knowledge in the pretrained VFMs.

4) Ablation for Injection Method: Different injection strategies in the Injection-Net may affect the effectiveness of contextual cues propagation. As shown in Table X, we experiment with various injection methods to investigate their performance differences in information transfer. In particular, we replace deformable attention [57] with addition, concatenation, cross attention [36], and FreqFusion [64] for comparison. In addition to concatenation and FreqFusion, we crop the region in the context feature to spatially align with the local feature. The results show that deformable attention achieves significantly superior performance under comparable numbers of trainable parameters in the backbone.

5) Ablation for Context Resolution: The context patch provides contextual information that enhances local semantic discrimination. We investigate the impact of the spatial resolution $H_C \times W_C$ of the context patch on segmentation performance in Table XI(a). The results show that the optimal performance is achieved when the spatial resolution is set to 1024×1024. The smaller resolution may provide insufficient contextual cues, whereas the overly large ones may suffer from information loss due to image downsampling. Nevertheless, all configurations with context outperform the model without contextual information.

6) Ablation for Context-Net Backbone: The Context-Net aims to effectively expand the receptive field of the Local-Net by capturing semantic information from context patches. To investigate the impact of the Context-Net's scale on the overall framework performance, we replace the Context-Net backbone from ViT-Base to ViT-Small. As shown in Table XI(b), this results in a significant drop in mIoU. This observation indicates that, although context features may introduce a certain degree of redundancy, a powerful feature representation capability remains indispensable for the Context-Net.

7) Ablation for Selection Strategies of CSAS: CSAS leverages the correctly predicted regions from Context-Net to enhance Local-Net́s sensitivity to large-scale areas. We also attempt to directly use pseudolabel $\hat{\mathbf{Y}}_C$ to provide additional supervision for the Local-Net. As shown in Table XII(a), the model performance drops significantly due to the noisy labels within $\hat{\mathbf{Y}}_C$. Furthermore, we investigate an alternative strategy that selects mispredicted regions of the Context-Net from local ground truth $\mathbf{Y}_P$ for additional supervision. This involves employing the XOR operation to derive the mask $\mathbf{M}$. The positions in the mask with values of 1 indicate regions where the Context-Net makes incorrect predictions. These regions typically contain more fine-grained details. The results reveal that using the XOR operation actually leads to negative effects. The primary reason may be that Local-Net focuses excessively on local details while neglecting large-scale regions.

8) Ablation for Hyperparameter β: Hyperparameter β in the loss function aims to adjust the sensitivity of Local-Net to large-scale regions. As shown in Table XII(b), we investigate the influence of the hyperparameter β. The results demonstrate that the segmentation performance gradually improves as β increases from 0.5 to 2. However, performance declines for larger β, possibly due to neglecting small-scale objects. Thus, we set $\beta = 2$.

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TABLE XIII
ABLATION STUDY ON DIFFERENT DECODERS. PARAM† REPRESENTS THE PARAMETERS OF THE DECODER. THE VALUES IN BOLD INDICATE THE BEST RESULTS

Decoder	Param†(M)	mIoU(%)
All-MLP [37]	4.7	70.1
UperNet [59]	32.3	**72.1**

9) Ablation for Different Decoders: The size of the decoder also determines the efficiency of parameter storage. Therefore, we adopt a lightweight All-MLP decoder [37] to investigate its impact on model performance. As shown in Table XIII, the All-MLP decoder has fewer parameters. Although the mIoU decreases, it remains above 70% mIoU.

VI. CONCLUSION

In this article, we propose the PEACE framework to effectively leverage pretrained knowledge from VFMs with only a minimal number of trainable parameters, enabling collaborative modeling of both local and corresponding contextual patches from UHR remote sensing images. In particular, we equip VFMs with the MRLA and MRCA to capture local and contextual spatial features via multiple receptive fields, respectively, which are then effectively coordinated through an injection network. Furthermore, CSAS encourages the model to focus on large-scale semantic regions. Experiments demonstrate that PEACE substantially reduces trainable parameters, exhibits strong versatility across various VFMs and RSFMs, and achieves state-of-the-art performance on multiple UHR remote sensing image benchmarks.

Limitations and Future Work: Although PEACE introduces only a small number of trainable parameters compared to the baseline, the inference speed of VFMs on UHR remote sensing images remains a significant challenge overall. Future work will focus on lightweighting the VFM and leveraging parallel computing strategies to further reduce memory and computational overhead, thereby facilitating large-scale applications.

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